

Layer 0 Diagnostic — Further Pure, Paper 1

Cold. No notes. ~90 minutes. Calculator allowed, but show every step — arithmetic slips at boundary/scaling steps are a logged gap and the working is what gets marked.

Before you start: write the start time. Record the stop time. Keep every piece of rough work, **including approaches you abandon** — the abandoned routes are the most informative thing in the packet.

Before you mark: for each question write your prediction — **SOLID / RUSTY / GAP** — *before* checking anything. The mismatch between prediction and result is the metacognition reading and it is the best-evidenced measurement in the whole system. Predictions written after the fact are worthless.

If a topic has not been taught yet, write "NOT TAUGHT" and move on. That is data, not failure — it separates "gap" from "not yet covered", and those need completely different responses.

Do not read §B until you have finished and marked.

§A The paper

Questions are interleaved — consecutive questions deliberately use different methods, and none tells you which method to use. That is the point.

1. Let $z = 1 - i\sqrt{3}$.

Find $|z|$ and $\arg z$, and hence evaluate z^6 .

2. $\mathbf{M} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$

Find $\mathbf{M}^{-1}$.

State clearly, before you begin, the condition that guarantees the inverse exists, and verify it holds.

3. The equation $x^3 - 5x^2 + 2x + 8 = 0$ has roots α, β, γ .

Write down $\sum \alpha$, $\sum \alpha\beta$ and $\alpha\beta\gamma$, and hence find $\sum \alpha^2$.

4. Two lines are given by

$$\ell_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \quad \ell_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

Determine whether the lines are parallel, intersecting or skew, **justifying your classification**, and find the shortest distance between them.

5. Prove by induction that $7^n + 2$ is divisible by 3 for all integers $n \geq 1$.

Before writing the inductive step, state **in words** exactly which statement you are assuming and exactly which statement you are proving. Then write the proof.

6. Sketch the locus of points in the Argand diagram satisfying

$$|z - 3 + 2i| = |z - 1|$$

and find its Cartesian equation.

7. The plane Π passes through $A(1, 2, -1)$, $B(2, 0, 3)$ and $C(0, 1, 1)$.

Find the perpendicular distance from $P(3, 3, 3)$ to Π .

8. Using the same cubic as question 3 — $x^3 - 5x^2 + 2x + 8 = 0$ with roots α, β, γ — find a cubic equation whose roots are $\alpha^2, \beta^2, \gamma^2$.

Do this **without** finding α, β, γ themselves.

9. Use standard results to find $\sum_{r=1}^n r(r+1)$, simplifying fully.

10. $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$

Find the equations of all lines through the origin that are invariant under the transformation represented by $\mathbf{A}$.

11. Use de Moivre's theorem to show that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

12. Find $\sum_{r=1}^n \frac{1}{r(r+2)}$ in terms of n , simplifying fully.

State what the sum tends to as $n \rightarrow \infty$.

13. $\mathbf{B} = \begin{pmatrix} 4 & 3 \\ 5 & 4 \end{pmatrix}$

Find $\mathbf{B}^{-1}$.

Then, in one sentence each: what does it mean geometrically for a 2×2 matrix to have determinant zero, and what does a **negative** determinant tell you about the transformation?

14. **Method selection. Do not solve any of these.** For each, state in one line which method you would use, **why that one rather than the obvious alternative**, and any condition that must hold first.

(a) Find the shortest distance from the point $(2, 1, 5)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 2, 2)$.

(b) Evaluate $\sum_{r=1}^{20} \frac{1}{(2r-1)(2r+1)}$.

(c) Solve $\mathbf{M}\mathbf{x} = \mathbf{b}$ for a 3×3 matrix $\mathbf{M}$ and a single given $\mathbf{b}$.

(d) Find $\alpha^3 + \beta^3 + \gamma^3$ for the roots of a given cubic.

15. Using your answer to question 10, find $\mathbf{A}^5$ **without** computing repeated matrix products.

16. **Conditions and definitions.** One or two sentences each, from memory.

(a) State the condition for a 3×3 matrix to represent an invertible transformation, and say what it means geometrically when it fails.

(b) Define what it means for two lines in three dimensions to be *skew*.

(c) In a proof by induction, what exactly is the base case doing? Why is the proof invalid without it?

(d) State the validity condition for the general binomial expansion of $(1+x)^n$ when n is not a positive integer.

(e) What is an eigenvector, in words, without using the word "eigenvalue"?

(f) Give the conditions under which $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ can fail.

END OF PAPER. Record the stop time. Write your SOLID / RUSTY / GAP predictions now, before reading on.

